TASK 3: REQUIREMENT ANALYSIS.

DESIGN AND IMPLEMENTATION OF A CROWDSENSED DRIVE TEST MOBILE APPLICATION.

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1. Introduction

This document presents Task 3 of the Crowdsensed Drive Test Mobile Application project, the Requirement Analysis phase. Building directly on the requirements gathered in Task 2, this task involves a systematic review of those requirements for completeness, clarity, technical feasibility, and dependency relationships. It identifies inconsistencies and ambiguities, classifies requirements into functional and non-functional categories, prioritises them using the MoSCoW framework, and produces the formal Software Requirements Specification (SRS).

Requirement analysis is a critical bridge between the informal outputs of the gathering phase and the formal specification that guides system design and implementation. A poorly analysed set of requirements leads to scope creep, missed features, and costly redesigns late in the development lifecycle. This analysis was conducted against the requirements document produced in Task 2 and cross-referenced with the drive test operational data, RF metric specifications, and user reluctance findings gathered during that phase.

2. Completeness, Clarity, Technical Feasibility, and Dependency Analysis

Each requirement area was evaluated against four criteria: completeness (are all necessary details present?), clarity (is the requirement unambiguous and testable?), technical feasibility (can it be implemented with available technology within the project constraints?), and dependency relationships (what other requirements does it depend on?).

Requirement Area	Completeness	Clarity	Feasibility	Status
RF measurement data collection	High - RSRP, RSRQ, RSSI, SINR, PCI, EARFCN all specified with example values	High -specific parameters with units defined	High - available via Android TelephonyManager API	Complete
GPS and location data	High - latitude, longitude, altitude, accuracy, speed, bearing, timestamp all listed	High - WGS84 standard specified	High - standard platform GPS API on both Android and iOS	Complete
Drive test session operations	High - vehicle setup, session duration, call drop protocol, Harvard sentences all documented	High - operational steps clearly described	High - well-understood operational process	Complete
Stakeholder roles	High - 8 roles with responsibilities documented	High - each role clearly distinguished	N/A	Complete
User reluctance and privacy	High - privacy, storage, trust, offline, ease of use all captured	High - direct user quotes support each concern	High - standard GDPR and encryption practices	Complete
Backend platform architecture	Low - no API spec, no database schema, no prediction algorithm	Low - 'backend platform' mentioned but not described	Medium - feasible but undefined	Incomplete
Mobile app UX and screens	Low - one-handed use	Low - no specific	High - feasible once specified	Incomplete

Requirement Area	Completeness	Clarity	Feasibility	Status
	implied but no wireframes or screen flows	interaction model described		
Security and data privacy	Medium - GDPR mentioned but no encryption standard, retention period, or consent flow	Low - too vague to be testable	High - standard TLS/AES implementation	Incomplete
5G NR metric support	Low - document references 2G/3G/4G only despite earlier scope including 5G	Low - ambiguous whether 5G is in or out of scope	High - available via Android APIs for 5G devices	Ambiguous
L3 message capture	Low - mentioned without definition of feasibility or method	Low - no specification of diagnostic access approach	Low - not accessible on standard consumer smartphones	Ambiguous

3. Inconsistencies, Ambiguities, and Missing Information

3.1 Inconsistencies

An inconsistency is a contradiction between two or more requirements or between a requirement and stated project goals. Three significant inconsistencies were identified:

Resolution required: The team must explicitly define whether the application targets (a) Professional Rig Mode integrating with existing vehicle rigs as a software layer, (b) Crowdsensed Smartphone Mode using everyday smartphones as standalone IoT collectors, or (c) both as distinct operating modes within the same application. This decision affects the data collection API design, the session management model, and the hardware assumptions embedded in the NFRs.

Resolution required: Confirm whether 5G NR is in scope for the first release. If yes, add SS-RSRP, SS-SINR, NR-ARFCN, gNodeB ID, and NR band to the measurement specification. If deferred to v2, document this explicitly with a rationale.

Resolution required: Confirm whether VoLTE/IMS call quality testing is in scope. If yes, specify the relevant SIP methods (INVITE, TRYING, RINGING, OK, BYE, CANCEL) and define how they relate to call setup time, call drop detection, and MOS scoring requirements.

3.2 Ambiguities

An ambiguity is a requirement that can be interpreted in multiple ways, leading to inconsistent implementation decisions.

ID	Ambiguity	Resolution
A-1	'Simulate certain other information related to CQI', unclear whether CQI should be read from the device hardware or synthetically generated	CQI is a real-time UE measurement available via Android TelephonyManager. Specify: CQI must be read from the device in real time, not simulated
A-2	'Coverage will get changed with time', vague. Could mean interference prediction, handover analysis, temporal coverage degradation, or all three	Define specifically: capture (a) handover events with timestamps, (b) RSRP trend over the drive route, (c) time-series coverage change for backend prediction models
A-3	L3 message analysis mentioned but standard consumer smartphones do not expose L3 messages via public APIs	Resolve: L3 capture is restricted to professional Android devices with AT command or diagnostic port access. Clarify whether this is a Phase 1 requirement or deferred
A-4	'How long will it take?' from user reluctance, ambiguous: refers to session duration, upload time, or dashboard results latency?	Address all three: session duration target ≤90 min; upload time target ≤30 seconds for a full session; dashboard refresh target ≤5 seconds after upload

3.3 Missing Requirements

The following required information was absent from the gathered requirements and must be added before the SRS can be finalised:

ID	Missing Requirement	Priority
M-1	No offline-first buffering specification despite users explicitly asking 'Can the app operate offline?' - buffer capacity, storage format, and upload trigger conditions are undefined	Critical
M-2	No data transmission protocol defined - REST, MQTT, or WebSocket? No API payload format specified - JSON, Protobuf? No API versioning strategy	Critical
M-3	No call drop detection requirement in the mobile app despite the vehicle rig using an audio alarm - threshold, alert type, and logging behaviour undefined	High
M-4	No coverage prediction algorithm specified for the backend - Kriging interpolation, Inverse Distance Weighting (IDW), or ML-based model? This is the core backend intelligence	High
M-5	No performance benchmarks - measurement sampling rate, maximum upload latency, minimum dashboard refresh rate, and concurrent device support all undefined	High
M-6	No multi-device session management requirement - crowdsensing implies many concurrent devices but there is no specification for device registration, session correlation, or data deduplication	Medium

4. Requirement Classification: Functional and Non-Functional

4.1 Functional Requirements

Functional requirements define what the system must do, the specific behaviours, functions, and capabilities that the application must exhibit. Twelve functional requirements were identified and specified:

FR-01	Collect RSRP, RSRQ, RSSI, SINR, CQI, EARFCN, PCI, Cell ID, MCC/MNC, and RAT type in real time during an active drive test session

FR-02	Record GPS coordinates (WGS84), altitude, GPS accuracy, speed, bearing, and UTC timestamp for every measurement sample

FR-03	Capture and log Layer 3 (L3) signalling messages and SIP protocol events where device diagnostic access permits (professional Android devices only)

FR-04	Support voice session testing using phonetically balanced Harvard sentences; log MOS score, call setup time, and call drop events

FR-05	Perform DL/UL throughput speed tests and log results with associated RF measurements and GPS position

FR-06	Detect call drop events during active sessions and trigger an audible alarm and visual alert to the operator

FR-07	Buffer all measurement samples locally when network connectivity is unavailable and automatically transmit to the backend upon reconnection

FR-08	Backend platform shall ingest measurement payloads from multiple concurrent devices and store with session ID, device ID, and operator metadata

FR-09	Backend shall generate a coverage heatmap using spatial interpolation of sample points, visualised on a geographic map layer

FR-10	Backend shall automatically flag coverage holes where RSRP falls below a configurable threshold (default: −110 dBm) and generate alerts

FR-11	Dashboard shall allow RF planners and NOC teams to export drive test session data as CSV files and formatted PDF reports

FR-12	Application shall anonymise device identifiers and user IDs before transmission in compliance with GDPR and applicable data protection legislation

4.2 Non-Functional Requirements

Non-functional requirements define how the system must perform, the quality attributes, constraints, and standards that govern its operation. Ten non-functional requirements were identified:

ID	Category	Requirement	Rationale
NFR-01	Performance	Collect measurements at a minimum rate of 1 sample per second during an active session	1 Hz sampling rate standard for drive testing; lower rates miss transient coverage events
NFR-02	Availability	Operate fully offline for a minimum of 90 minutes without data loss	Drive test sessions run up to 90 minutes in areas with no data connectivity
NFR-03	Power efficiency	Consume no more than 15% device battery per hour during active collection	Field sessions may run 4–6 hours; excessive drain risks session failure mid-route
NFR-04	Storage	Installed application size shall not exceed 50 MB	User concern: 'I don't want it taking too much space in my phone'
NFR-05	API performance	Backend API shall respond to measurement ingestion requests within 500 ms under normal load	Ensures upload does not block the main thread or delay subsequent samples
NFR-06	Dashboard	Coverage map shall refresh within 5 seconds of a new batch upload completion	RF planners need near-real-time visibility during active multi-device sessions
NFR-07	Security	All data in transit shall use TLS 1.3 minimum; locally buffered samples shall use AES-256 at rest	User concern: data exposure; GDPR compliance for location data
NFR-08	Usability	Application UI shall be fully operable with one hand on screens between 5 and 7 inches in direct sunlight (minimum contrast ratio 4.5:1 per WCAG 2.1 AA)	Field technicians operate the app while in moving vehicles; sunlight glare is a real operational condition
NFR-09	Scalability	System shall support a minimum of 50 concurrent device uploads without degraded API ingestion performance	Crowdsensing model requires simultaneous multi-device operation across a city

ID	Category	Requirement	Rationale
NFR-10	Compatibility	Application shall support Android 10 (API level 29) and above, and iOS 14 and above	Covers the majority of active field devices; TelephonyManager APIs required for measurement access

5. Requirement Prioritisation

Requirements were prioritised using the MoSCoW framework; Must Have, Should Have, Could Have, and Won't Have. Priority was assigned based on two dimensions: business importance (impact on the app's core value proposition) and technical feasibility (complexity and risk of implementation within the project timeline).

5.1 Must Have:

These requirements represent the minimum viable product. The application cannot be considered functional without them:

ID	Requirement	Justification
FR-01	Real-time RF measurement collection	Core app function - without this the app has no purpose
FR-02	GPS stamping of every sample	Coverage maps are meaningless without location context
FR-07	Offline buffering and auto-upload	Drive tests occur in coverage-weak areas - offline is not optional
FR-08	Backend measurement ingestion API	Without ingestion there is no platform - just a local logging app
FR-10	Automated coverage hole detection	Primary analytical output of the platform - core deliverable
NFR-02	90-minute offline session	Matches real drive test session duration requirements
NFR-07	TLS 1.3 and AES-256 encryption	GDPR compliance is a legal obligation, not a feature

5.2 Should Have:

These requirements significantly enhance the value of the application and should be included in the first production release if timeline permits:

ID	Requirement	Justification
FR-05	DL/UL throughput testing	Key QoS metric for data performance assessment
FR-06	Call drop detection and alert	Critical for session integrity - mirrors the vehicle rig audio alarm
FR-09	Coverage heatmap with interpolation	Core visualisation for RF planners on the dashboard
FR-11	CSV/PDF export	Required for post-session reporting and engineering decision-making

ID	Requirement	Justification
NFR-03	Battery ≤15%/hour	Field sessions last several hours - battery management is essential
NFR-08	One-handed sunlight-readable UI	Primary use context - in-vehicle, one-handed, outdoor

5.3 Could Have:

ID	Requirement	Reason Deferred
FR-03	L3 message capture	Requires rooted device or manufacturer diagnostic agreement - high technical risk
FR-04	Harvard sentence voice test + MOS	Requires additional audio processing infrastructure and VoLTE scope confirmation
NFR-09	50+ concurrent device support	Infrastructure scaling - can be addressed after initial deployment

5.4 Won't Have:

Item	Reason
In-building coverage measurement	Requires specialist IBE engineers and DAS equipment - separate project
Multi-operator comparative benchmarking	Requires simultaneous multi-SIM measurement - hardware complexity out of scope
Real-time ML prediction on-device	Device compute constraints; model training requires backend infrastructure first

6. Dependency Relationships

Understanding how requirements depend on each other is essential for planning the implementation sequence and identifying risks. The following dependency chain was identified:

Requirement	Depends On	Type of Dependency
FR-09 (Coverage heatmap)	FR-02 (GPS stamping)	Hard - heatmap requires geo-referenced samples
FR-09 (Coverage heatmap)	FR-01 (RF measurement)	Hard - heatmap requires RSRP values to plot signal strength
FR-10 (Hole detection)	FR-01 (RF measurement)	Hard - hole detection requires RSRP threshold comparison
FR-10 (Hole detection)	FR-09 (Heatmap)	Soft - hole detection is more meaningful with interpolated heatmap context
FR-08 (Backend ingestion)	FR-07 (Offline buffer)	Hard - ingestion must accept batched uploads from offline buffer
FR-07 (Offline buffer)	NFR-02 (90-min capacity)	Hard - buffer must hold a minimum of 90 minutes of samples
FR-05 (Throughput test)	FR-02 (GPS stamping)	Hard - throughput results must be geo-referenced
FR-06 (Call drop alert)	FR-04 (Voice session)	Hard - call drop detection only applies during active voice sessions
FR-12 (Privacy - anonymisation)	All FRs involving data transmission	Cross-cutting - anonymisation applies at every data collection and upload point
NFR-10 (Android 10+ / iOS 14+)	FR-01 (RF measurement)	Hard - TelephonyManager APIs for RSRP/RSRQ require Android 10 minimum

7. Software Requirements Specification (SRS)

7.1 Document Purpose

This Software Requirements Specification documents the agreed requirements for the Crowdsensed Drive Test Mobile Application. It serves as the contractual basis between the development team and project stakeholders, defining what the system will do (functional requirements), how well it will do it (non-functional requirements), and the constraints within which it must operate.

7.2 System Overview

The system comprises three components:

- Mobile Collector App; an Android and iOS application installed on smartphones that collects cellular network measurements, records GPS position, buffers data locally, and transmits to the backend. Operates as a Class-2 IoT device.
- Backend Ingestion and Analytics Platform ; a cloud-hosted API that receives measurement payloads from multiple concurrent devices, stores them in a structured database, processes them for coverage analysis, and serves data to the dashboard.
- Web-Based Analytics Dashboard; a browser-accessible interface for RF planners, NOC teams, and executives to view live coverage heatmaps, coverage hole alerts, session history, and exportable reports.

7.3 Scope

In scope for this release: Android 10+ and iOS 14+ mobile collection; 2G, 3G, and 4G LTE RF measurements; GPS-referenced drive test sessions; offline buffering; backend ingestion; coverage heatmap; coverage hole detection; dashboard visualisation; CSV and PDF export; GDPR-compliant data handling.

Out of scope for this release: 5G NR measurements (pending confirmation); L3 message capture; VoLTE/Harvard sentence voice testing; in-building coverage; multi-operator benchmarking; on-device ML prediction.

7.4 Definitions and Acronyms

Term	Definition
RSRP	Reference Signal Received Power (dBm) ; primary LTE/5G coverage strength indicator
RSRQ	Reference Signal Received Quality (dB) ; signal quality relative to interference level
RSSI	Received Signal Strength Indicator (dBm) ; total received wideband power

Term	Definition
SINR	Signal to Interference plus Noise Ratio (dB) ; channel quality metric
EARFCN	E-UTRA Absolute Radio Frequency Channel Number ; identifies the LTE carrier frequency
PCI	Physical Cell Identity ; unique identifier for a cell within a network area
L3	Layer 3 ; network layer signalling (RRC, NAS messages in LTE/5G)
MOS	Mean Opinion Score ; perceptual voice quality rating (1–5 scale)
SBC	Single Board Computer ; embedded controller used in professional drive test rigs
RAT	Radio Access Technology ; 2G GSM, 3G UMTS/WCDMA, 4G LTE, 5G NR
MNO	Mobile Network Operator ; the telecommunications company operating the network under test
IoT	Internet of Things ; the smartphone acts as a Class-2 IoT device in this context
GDPR	General Data Protection Regulation ; EU data protection framework applied to user location data
MoSCoW	Must Have / Should Have / Could Have / Won't Have ; requirement prioritisation framework

7.5 Constraints and Assumptions

Technical Constraints

- iOS does not expose RSRP, RSRQ, PCI, or EARFCN to third-party applications via public APIs. The iOS build will be limited to network type, throughput, latency, and GPS. This is a platform-imposed constraint that cannot be resolved without Apple developer entitlements.
- L3 message access on Android requires either root access or a manufacturer-specific diagnostic interface agreement. Standard consumer smartphones do not expose L3 via TelephonyManager.
- GPS accuracy degrades in urban canyons and tunnels. Samples with GPS accuracy below 50 metres should be flagged rather than discarded, as even imprecise location data contributes to coverage models.

Operational Constraints

- The application must not require any modification to existing drive test vehicle hardware or software in Professional Rig Mode.
- Session data must remain accessible and exportable even if the backend platform is temporarily unavailable.
- The application must comply with GDPR and any applicable national data protection legislation in the deployment country.

Assumptions

- Target devices have a minimum of 2 GB RAM and 16 GB storage to support the local measurement buffer.
- The backend will be deployed on a cloud platform (AWS, GCP, or Azure) with auto-scaling capabilities.
- Users will have accepted the application's privacy policy and granted all required permissions (location, cellular state, storage) before initiating a session.

7.6 Complete Functional Requirements Specification

ID	Description	Priority	Depends On	Verification Method
FR-01	Collect RSRP, RSRQ, RSSI, SINR, CQI, EARFCN, PCI, Cell ID, MCC/MNC, RAT type at ≥1 Hz during active session	Must	NFR-10	Automated test: verify all fields populated in sample JSON
FR-02	Record GPS lat/lon, altitude, accuracy, speed, bearing, UTC timestamp per sample	Must	None	Automated test: verify GPS fields within expected ranges
FR-03	Capture L3 signalling messages on diagnostic-enabled Android devices	Could	FR-01	Manual test: verify L3 logs on rooted test device
FR-04	Voice session testing with Harvard sentences; log MOS, setup time, drop events	Could	FR-06	Manual test: verify MOS calculation against reference audio
FR-05	DL/UL throughput test with RF context and GPS	Should	FR-01, FR-02	Automated test: verify throughput logged with associated RSRP
FR-06	Detect call drop and trigger audible + visual alert	Should	None	Manual test: simulate call drop, verify alert fires
FR-07	Buffer samples locally offline; auto-transmit on reconnection	Must	NFR-02	Test: collect samples offline, verify upload on reconnect
FR-08	Backend ingests payloads from ≥50 concurrent devices	Must	FR-07	Load test: simulate 50 concurrent device uploads
FR-09	Generate RSRP coverage heatmap with spatial interpolation	Should	FR-01, FR-02	Visual inspection: verify heatmap matches sample distribution
FR-10	Flag coverage holes where RSRP < configurable threshold (default −110 dBm)	Must	FR-01	Automated test: inject below-threshold sample, verify alert
FR-11	Export session data as CSV and PDF from dashboard	Should	FR-08	Manual test: trigger export, verify file format and content

ID	Description	Priority	Depends On	Verification Method
FR-12	Anonymise device/user IDs before transmission; GDPR compliant	Must	All FRs	Code review: verify no PII in transmitted payload

7.7 Complete Non-Functional Requirements Specification

ID	Category	Requirement	Metric	Verification
NFR-01	Performance	Sample collection rate	≥1 sample/second	Automated timing test
NFR-02	Availability	Offline operation duration	≥90 minutes without data loss	Field test: disable network, run session
NFR-03	Power	Battery consumption	≤15% per hour	Measure battery drain over 1-hour session
NFR-04	Storage	App installed size	≤50 MB	Build artifact size check
NFR-05	API performance	Ingestion response time	≤500 ms per request	Load test with simulated payloads
NFR-06	Dashboard	Map refresh latency	≤5 seconds post-upload	Measure end-to-end from upload to map update
NFR-07	Security	Encryption standard	TLS 1.3 in transit; AES-256 at rest	Penetration test; code review
NFR-08	Usability	One-handed operation; contrast ratio	Operable 5–7 inch screen; ≥4.5:1 contrast	Usability test with target users
NFR-09	Scalability	Concurrent device support	≥50 concurrent uploads	Cloud load test
NFR-10	Compatibility	Platform support	Android 10+; iOS 14+	Device matrix testing

8. Stakeholder Validation

Requirement validation ensures that the documented requirements accurately represent stakeholder needs and are consistent, complete, and achievable. The following checklist documents the current validation status and identifies items requiring stakeholder sign-off before development can begin.

Validation Item	Responsible Stakeholder	Status	Action Required
Drive test technician confirmed operational workflow	Drive Test Technician	Confirmed	None
Offline session duration validated against storage estimates	Drive Test Technician / Engineer	Partial	Calculate storage for 90-min × 1Hz × sample size
MNO secondary research validated as sufficient substitute	Supervisor / RF Engineer	Partial	Supervisor sign-off on secondary research approach
All 8 stakeholder roles reviewed and approved	All stakeholders	Partial	Formal sign-off session required
RF planner confirmed dashboard KPI list	Network Planning Engineer	Outstanding	Schedule stakeholder review session
Data scientist confirmed prediction algorithm	Data Scientist	Outstanding	Select algorithm (Kriging/IDW/ML) and document
Data privacy advocate reviewed anonymisation approach	Data Privacy Advocate	Outstanding	Draft anonymisation policy for review
NOC team confirmed alert thresholds	NOC Team	Outstanding	Define RSRP threshold and notification channels
5G NR scope confirmed or deferred	Project Team	Outstanding	Team decision required - document in SRS
L3 capture feasibility confirmed	Technical Team	Outstanding	Assess diagnostic API access on target devices
iOS API limitations acknowledged and scope adjusted	Technical Team	Outstanding	Document iOS-specific feature limitations
SIP message types defined and VoLTE scope confirmed	RF / Technical Team	Outstanding	Confirm VoLTE in or out of scope
All ambiguities (A-1 to A-4) resolved	Project Team + Supervisor	Outstanding	Resolution meeting required
Requirements baselined and version-controlled	Project Team	Outstanding	Commit to version control after sign-offs complete